

Tech Stack & Architecture: RestroPOS

This document provides a non-developer-friendly breakdown of the technical stack, database model, and server architecture powering RestroPOS.

1. High-Level Architecture

RestroPOS is built using a **serverless, multi-tenant database-per-restaurant model** that guarantees absolute security (Row-Level Security) and fast page load times.

```
graph TD
  subgraph Client ["Browser / Device"]
    UI["React & Tailwind UI"]
    AuthClient["Neon Auth Client"]
  end

  subgraph Server ["Vercel Serverless Edge"]
    API["Next.js API Routes"]
    Prisma["Prisma ORM (Tenant Filter)"]
    Middleware["Next.js Router Middleware"]
  end

  subgraph Database ["Neon Postgres Database"]
    SchemaAuth["neon_auth Schema (User Accounts & Sessions)"]
    SchemaPublic["public Schema (Tables, Menu, Orders, Inventory)"]
  end

  UI -->|1. Sign In / Request| Middleware
  Middleware -->|2. Verify Cookie| SchemaAuth
  API -->|3. Query via Prisma| Prisma
  Prisma -->|4. SQL Execution (Scoped)| SchemaPublic
```

2. The Tech Stack Explained

We selected a modern, premium stack designed for scalability, zero maintenance, and instant performance:

■ Frontend & Application Layer

1. **Next.js 16 (App Router):** The foundation of the app. It handles both building the user interface (frontend) and executing backend logic (Server Components & API routes) in the same codebase.

2. **React 19:** Powers the interactive user interface (e.g., live cart calculation, real-time table statuses, and interactive analytics charts).
3. **TailwindCSS v4:** Utility-first styling framework used to build the premium, dark-mode-first aesthetic without writing bloated custom CSS.
4. **Lucide React:** A clean icon library providing consistent, minimalist visual indicators for navigation and actions.

■ Authentication & Security Layer

1. **Neon Auth (Powered by Better Auth):** Neon's managed authentication service. It stores credentials securely, sends verification emails/OTPs, handles password resets, and manages secure session cookies (`__Secure-neonauth.session_token`) natively inside the database.
2. **Prisma ORM & Client Extensions:** A database query builder. We added custom middleware that automatically appends the user's `restaurantId` to *every single query*, making it impossible for cashier A at restaurant A to see table configurations or orders from restaurant B.

■ Database Layer (The Engine)

1. **Neon Serverless Postgres:** A cloud database that automatically scales compute resources up during busy dinner rushes and scales down to zero when the restaurant is closed to save costs.
2. **Connection Pooling:** Ensures that hundreds of tables scanning QR codes and placing orders simultaneously do not overload the database connections.

3. Database Schema Blueprint

Here is how data is structured and linked inside the database (`schema.prisma`):

Model / Table | Purpose | Relationships

:--- | :--- | :---

Restaurant | The root tenant. Stores name, phone, currency, GSTIN, and settings. | Has many `User`, `Table`, `Category`, `Order`, `InventoryItem`.

User | Store accounts of POS users (Cashier, Owner, Manager). | Belongs to `Restaurant`, has many `Order`.

Table | Restaurant tables. Tracks status (`AVAILABLE`, `OCCUPIED`, etc.) and capacity. | Belongs to `Restaurant`, has many `Order` and `TableSession`.

Category | Menu groupings (e.g., "Starters", "Mains"). | Belongs to `Restaurant`, has many `MenuItem`.

MenuItem | Individual food items. Tracks price, veg/non-veg status, and tax. | Belongs to Category, has many Variant and OrderItem.

Order | Tracks sales. Stores subtotal, tax, discounts, status, and payment logs. | Belongs to Table, Restaurant, Customer.

TableSession | Ephemeral guest checkout sessions. Generates a secure UUID and a 4-digit PIN for steward validation to prevent URL tampering. | Belongs to Table and Restaurant.

InventoryItem | Raw materials (e.g., flour, milk). | Belongs to Restaurant, links to MenuItem for automatic auto-deduct.

StaffProfile | Tracks employees' shift types, salary details, and attendance logs. | Belongs to Restaurant.